CHEAT SHEET · DAY 4

Regression and statistical series

Tests, models, margins, the CPI as a time series and a panel. Every command here was taught today; the deck shows each one at work on the LFS or the CPI.

Tests — Part 1

`ci means y if emp == 1`	mean with 95% interval (no design)
`ttest y, by(g)`	two means, quick look
`prtest flag, by(g)`	two shares, quick look
`svy, subpop(s): mean y, over(g)`	design-based means by group
`mean, coeflegend`	the names of the estimates
`lincom c.y@1.g - c.y@2.g`	the gap with SE and interval
`test c.y@1.g = c.y@2.g`	the same as a p-value

Regression — Part 2

`regress y x if emp == 1`	linear regression
`i.region`	one coefficient per category
`ib7.region`	category 7 as the base
`c.age##c.age`	age and age squared
`predict yhat if e(sample)`	fitted values
`predict res, residuals`	residuals
`display e(N) e(r2)`	stored results

Diagnostics — Part 3

`rvfplot, yline(0)`	residuals against fitted
`estat hettest`	unequal spread? (plain regress)
`estat vif`	overlapping characteristics
`lvr2plot`	rows that steer the fit
`predict lev, leverage`	leverage per row
`regress ..., vce(cluster psu)`	clustered SE
`svy, subpop(emp): regress ...`	the SE NBS reports

Logit — Part 4

`logit flag i.x if lf == 1`	model for a 0/1 outcome
`logit ..., or`	odds ratios
`margins i.agegrp`	predicted rate per group
`margins, dydx(ilo_sex)`	difference in points
`marginsplot`	chart of the last margins
`svy, subpop(lf): logit ...`	design-based
`tabulate x flag`	check for empty cells first

Presenting — Part 5

`estimates store name`	keep a model
`estimates dir`	list stored models
`etable, estimates(a b)`	models side by side
`..., column(estimates) mstat(N)`	names and N
`..., export("f.docx", replace)`	to Word or Excel
`putdocx begin / text / image / save`	a note from the do-file
`graph export "f.png", replace`	a chart for the note

Time series — Part 6

`tsset month, monthly`	declare a monthly series
`tsline cpi`	chart it
`L.x L12.x D.x S12.x`	last month, last year, changes
`generate yoy = 100*(cpi/L12.cpi - 1)`	year-on-year inflation
`generate double s = S12.cpi`	keep the digits
`tssmooth ma m = x, window(11 1 0)`	12-month moving average
`tsreport`	gaps in the series

Panels — Part 7

`xtset division month, monthly`	declare a panel
`L12.cpi`	last year, same division
`if month == tm(2026m1)`	a monthly date in a condition
`xtline y, overlay`	one line per series
`xtsum y`	between and within
`gsort -y`	largest first

Further learning (not taught this week): multilevel models

Households within PSUs, PSUs within districts: when the question is about variation *between* levels, Stata's multilevel commands `mixed` (continuous outcome) and `melogit` (0/1 outcome) are the next step. Start with `help mixed` and the worked examples in the *Multilevel Mixed-Effects Reference Manual* linked from it. Design-based estimates for publication stay with `svy:`.

Help inside Stata: `help` *command*, and the PDF manual linked at the top of each help page.